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Experiment: 4

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4. Entropy, Cross-Entropy and KL Divergence

Scenario

A decision-support system classifies transactions into three risk categories. Calculate the entropy of the predicted class distribution to measure its uncertainty**.**

Algorithm

1. Start the program.
2. Import NumPy, Pandas, Matplotlib, and SciPy libraries.
3. Define the three transaction risk categories: Low Risk, Medium Risk, and High Risk.
4. Assign the predicted probabilities as 0.60, 0.30, and 0.10.
5. Define the actual probabilities as 0.50, 0.30, and 0.20.
6. Calculate the entropy of the predicted probability distribution.
7. Calculate the cross-entropy between the actual and predicted distributions.
8. Calculate the KL divergence between the actual and predicted distributions.
9. Create a summary table containing the risk categories and their predicted and actual probabilities.
10. Display the calculated entropy, cross-entropy, and KL divergence values.
11. Plot a bar chart to compare the predicted probabilities of the three risk categories.
12. Plot another bar chart to compare entropy, cross-entropy, and KL divergence.
13. Display the output and graphs.
14. Stop the program.

Source Code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
```

```
from scipy.stats import entropy

# 1. Define predicted probabilities for three risk categories
categories = ["Low Risk", "Medium Risk", "High Risk"]
predicted_probs = np.array([0.60, 0.30, 0.10])

# 2. Calculate entropy (in bits)
predicted_entropy = entropy(predicted_probs, base=2)

# 3. Define actual distribution for comparison
actual_probs = np.array([0.50, 0.30, 0.20])

# 4. Calculate cross-entropy (in bits)
cross_entropy = -np.sum(actual_probs * np.log2(predicted_probs))

# 5. Calculate KL divergence (in bits)
kl_divergence = entropy(actual_probs, predicted_probs, base=2)

# 6. Create summary table
summary = pd.DataFrame({
   "Risk Category": categories,
   "Predicted Probability": predicted_probs,
   "Actual Probability": actual_probs
})

print("TRANSACTION RISK DISTRIBUTION")
print(summary.to_string(index=False))

print("\nENTROPY ANALYSIS")
```

```
print(f"Predicted Entropy: {predicted_entropy:.4f} bits")
print(f"Cross-Entropy: {cross_entropy:.4f} bits")
print(f"KL Divergence: {kl_divergence:.4f} bits")

# 7. Plot predicted class distribution
plt.figure(figsize=(8, 5))
bars = plt.bar(categories, predicted_probs)

plt.title("Predicted Transaction Risk Distribution")
plt.xlabel("Risk Category")
plt.ylabel("Probability")
plt.ylim(0, 1)

for bar, value in zip(bars, predicted_probs):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.02,
        f"{value:.2f}",
        ha="center"
    )

plt.tight_layout()
plt.show()

# 8. Plot entropy, cross-entropy and KL divergence
metrics = ["Entropy", "Cross-Entropy", "KL Divergence"]
values = [predicted_entropy, cross_entropy, kl_divergence]

plt.figure(figsize=(8, 5))
```

```
bars = plt.bar(metrics, values)

plt.title("Entropy and Divergence Analysis")
plt.ylabel("Value (bits)")
plt.xticks(rotation=10)

for bar, value in zip(bars, values):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.02,
        f"{value:.4f}",
        ha="center"
    )

plt.tight_layout()
plt.show()
```

Output

TRANSACTION RISK DISTRIBUTION

Risk Category	Predicted Probability	Actual Probability
Low Risk	0.6	0.5
Medium Risk	0.3	0.3
High Risk	0.1	0.2

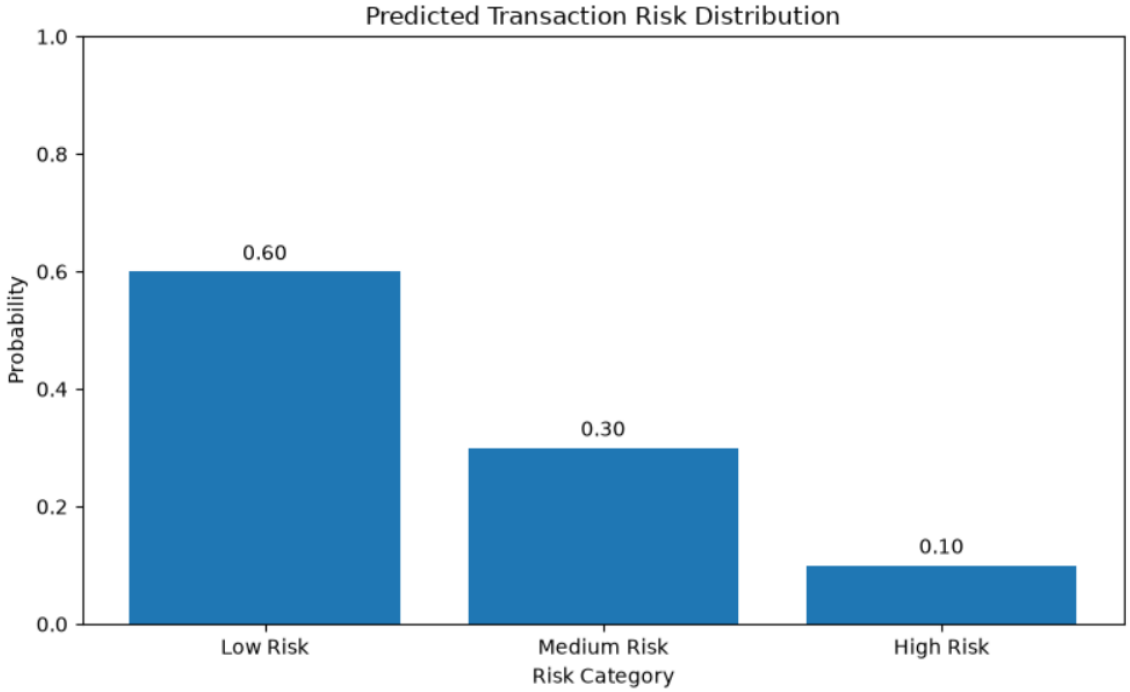
ENTROPY ANALYSIS

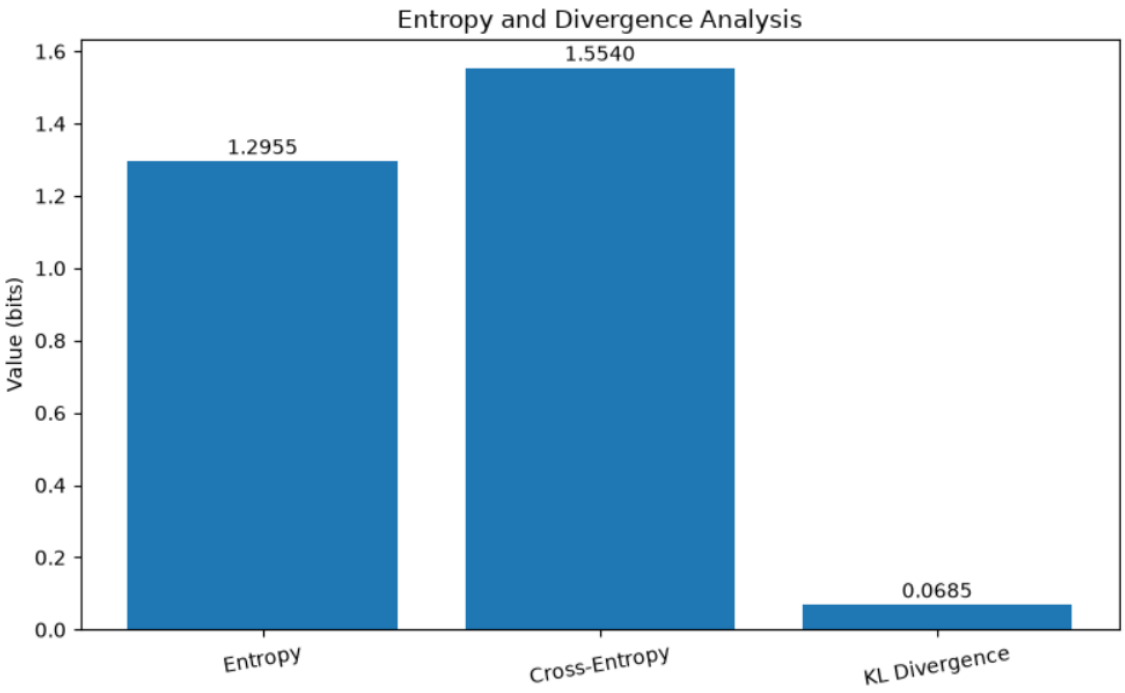
Predicted Entropy: 1.2955 bits

Cross-Entropy: 1.5540 bits

KL Divergence: 0.0685 bits

Cross-Entropy: 1.5540 bits
KL Divergence: 0.0685 bits




Entropy and Divergence Analysis
Value (bits)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.2955
1.5540
0.0685
Entropy
Cross-Entropy
KL Divergence